

Kirill Mazur

✉ k.mazur21@imperial.ac.uk | 🏠 makezur.github.io | 📄 Google Scholar

Education

Imperial College London

Ph.D. student in Robotic Vision

2021 - present

- Dyson Robotics Lab; Supervisor: [Andrew J. Davison](#)

Skolkovo Institute of Science and Technology, GPA: 3.83 out of 4.00

2018 - 2020

M.Sc. in Mathematics and Computer Science (with honors)

- Thesis: “Cloud Transformers”. Supervisors: [Victor Lempitsky](#) and [Gonzalo Ferrer](#)
- First year project: “Learnable point cloud descriptors for depth-only odometry and SLAM”

Yandex School of Data Analysis (YSDA), GPA: 4.84 out of 5.00

2016 - 2018

Computer Science (Masters-level degree school organized by Yandex)

Higher School of Economics (HSE), GPA: 9.22 out of 10.0

2014 - 2018

B.Sc. in Mathematics

- Thesis: “On the Families of Quartics and Rational Curves on the Quotient of the Quartic by the Involution”

Publications

4D Primitive-Mâché: Glueing Primitives for Persistent 4D Scene Reconstruction

[Kirill Mazur](#), Marwan Taher, Andrew J. Davison

[[project page](#), [video](#)]

Primitive-based 4D replayable reconstructions from casual videos.
preprint, 2025

KV-Tracker: Real-Time Pose Tracking with Transformers

Marwan Taher, Ignacio Alzugaray, [Kirill Mazur](#), Xin Kong, Andrew J. Davison

[[project page](#), [video](#)]

Real-time pose tracking from feed-forward reconstruction models.
preprint, 2025

SuperPrimitive: Scene Reconstruction at a Primitive Level

[Kirill Mazur](#), Gwangbin Bae, Andrew J. Davison

[[project page](#), [video](#), [code](#)]

Introduces a new representation for monocular scene reconstruction based on image segments.
(CVPR) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024

Honorable Mention at CVPR Demo Track

Feature-Realistic Neural Fusion for Real-Time, Open Set Scene Understanding

[Kirill Mazur](#), Edgar Sucar, Andrew J. Davison

[[project page](#), [video](#)]

Presents a new real-time high-dimensional feature fusion technique for open set scene understanding.
(ICRA) *IEEE International Conference on Robotics and Automation* 2023

Point-Based Clothing Modeling

Ilya Zakharkin*, [Kirill Mazur](#)*, Artur Grigorev, Victor Lempitsky

[[project page](#), [code](#)]

Presents a new technique for a visual try-on and clothes re-targeting for complex garments.
(ICCV) *IEEE/CVF International Conference on Computer Vision* 2021

Cloud Transformers: A Universal Approach To Point Cloud Processing Tasks

[Kirill Mazur](#), Victor Lempitsky

[[project page](#), [code](#)]

Presents a new layer for 3D point clouds processing that achieved SoTA results on four various tasks.
(ICCV) *IEEE/CVF International Conference on Computer Vision* 2021

Experience

Meta

Aug. 2024 - Feb. 2025

AI Research Scientist Intern at [Project Aria](#)

- Developed a model for dense metric geometry estimation for Aria device
- Large-scale training of dense point-map estimators, including data preparation and model iteration

Samsung AI Center Research Engineer at Virtual Human Telepresence Lab - Published 2 first (co-)author papers at a top computer vision venue (ICCV). “Cloud Transformers”: Research on neural point cloud processing. “Point-Based Clothing Modeling”: Research on neural 3D clothes modeling and its visual try-on.	Jun. 2019 - Jul. 2021
Yandex Intern ML Engineer at Computer Vision Lab - Developed a model for Object Localization which reduced the response time by 75% and doubled a target product metric - Adapted Tensorflow’s object detection project for Yandex Infrastructure	Jun. 2018 - Sep. 2018
Invited Talks	
Technical University of Munich (TUM) Week-long research visit to Daniel Cremers’ lab	July. 2024
Massachusetts Institute of Technology (MIT) “Always-On Single-Image Priors for Spatial AI”	Jun. 2024
Rank Symposium: Neural Rendering in Computer Vision “Feature-Realistic SLAM”	Aug. 2022
Academic Service	
Reviewer WACV 2025, ICCV 2025, CVPR 2024-2025, ICRA 2023-2024	
Imperial College London TA at Robotics course	Spring 2023, Spring 2024
Skolkovo Institute of Science and Technology TA at Deep Learning course	Spring 2020
Higher School of Economics / Yandex School of Data Analysis Research seminar curator	Sep. 2018 - Dec. 2019
Yandex School of Data Analysis (YSDA) TA at Algorithms and Data Structures course	Sep. 2018 - Jan. 2019